Exercise 1: Loyalty System on Ethereum

1. Fee and Operation Mechanism (Gas Fee)

When users transfer Loyalty Points to each other, they must pay a fee calculated in Gas.

The main purpose of the Gas charging mechanism is for network safety:
Gas is a unit of measurement for the computational effort required to execute an order in a Smart Contract. Gas fees are established with the important purpose of preventing infinite loops and spam attacks on the network. By requiring users to pay for each computational operation, the Gas mechanism compensates miners (or validators) and ensures that no one can use network resources indefinitely without limitation.

2. Distinguishing Account Roles The Ethereum system uses two main types of accounts:

Classification	Name (Role in the system)	Characteristics
Externally owned account (EOA)	User (Customer)	Private Key is required to sign Receive transactions and control accounts. Used to send Ether and initiate transactions.
Contract Account	The source code that defines the game rules (Smart Contract)	Controlled by source code (Code). No Private Key. Runs automatically when a transaction is received.

The account that is required to have a Private Key to confirm transactions is an Externally Owned Account (EOA), because this is how users control their assets and authorize transactions.

Exercise 2: Bank Alliance Network (Hyperledger Fabric) 1.

Permissioned Network model

Unlike Ethereum, which is a public network, Hyperledger Fabric operates on a Permissioned model.

If an unknown financial institution wants to participate in this network, they cannot do so freely. All members participating in the network must have their identities authenticated (Known Identities).

The component in the Fabric architecture responsible for managing identities and issuing digital certificates to members is the Membership Service Provider (MSP).

2. Privacy & Channels

The problem is that Bank C is a competitor, so A and B absolutely do not want C to see their transaction data.

Hyperledger Fabric solution provides Channels feature.

A and B need to create a private Channel, defined as a private subnet between only these two specific members.

When A and B execute separate credit agreements, the transaction is recorded only on that Channel's ledger.

This feature ensures that transactions in Channel A-B are completely invisible to non-Channel members (like Bank C), even though they are all on the same Hyperledger network.

3. Cost and Currency (No Cryptocurrency)

For the Fabric network of 3 banks, they do not need to issue or buy any virtual currency (Coin) to pay transaction or mining fees, because Fabric does not use Cryptocurrency.

This feature is very suitable for the business model (B2B) because:

1. It reduces the complexity of financial management and legal compliance, which is a major barrier when businesses use cryptocurrency.

2. Fabric is designed to serve a B2B (Business to Business) environment, where privacy, modular architecture flexibility and permissions are prioritized over the openness of Ethereum (which is more suitable for B2C/DApps).